CS24351 — ELECTIVE (3.0 CR) MODULE IV NOTES BIT MESRA | NEP SCHEME

Module 4: Neural Sequence Models & Transformers

RNN, LSTM, Seq2Seq Attention, Transformer Architecture, BERT, GPT

Module IV: Neural Sequence Models & Transformers — Topics Covered

1. Recurrent Neural Networks (RNN) & BPTT
2. Long Short-Term Memory (LSTM) & GRU Gating
3. Seq2Seq Architecture & Attention Mechanisms
4. The Transformer: Scaled Dot-Product & Multi-Head
5. Pretrained Models: BERT (MLM) vs. GPT (Autoregressive)

1. The Transformer Attention Formulation (Vaswani et al., 2017)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h) W^O \quad \text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

🏛 **BIT Mesra Exam Question (10 Marks)**

Problem: Compare BERT and GPT architectures in terms of attention directionality, pre-training objectives, and downstream fine-tuning.

Solution:

- **BERT:** Bidirectional Transformer Encoder. Pretrained on Masked Language Modeling (MLM 15% tokens) + Next Sentence Prediction (NSP). Ideal for classification, NER, QA extraction.
- **GPT:** Unidirectional Autoregressive Decoder (Causal attention mask). Pretrained on Next-Token Prediction. Ideal for text generation, creative writing, and in-context few-shot prompting.

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